

PROJECT REPORT

AI Career Guidance and Skill Recommendation System

No. of Students: 4

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Submitted By: Arya Singh Vats, Ruchi Sirohi, Akhilesh Chahar, Aryan Panwar

Submitted To: Ms Karuna Kaushik

1) Objective

The main objective of the AI Career Guidance and Skill Recommendation System is to help students and job seekers make informed career decisions by analyzing their interests, skills, academic background, career goals, and current skill gaps using Artificial Intelligence.

Objectives:

To provide personalized career recommendations based on an individual's profile, interests, and skills.

To identify skill gaps between the user's current capabilities and the requirements of target careers.

To recommend relevant technical and soft skills that can improve career readiness.

To suggest suitable learning paths, courses, certifications, and project areas.

To help users understand suitable career options and the skills required for them.

To provide data-driven guidance that supports better academic and professional decision-making.

2) Questionnaire Design

A questionnaire can be used to understand user requirements, career preferences, existing skills, learning needs, and expectations from an AI-based career guidance platform.

Sample Questionnaire

1. What is your current educational or professional background?

School Student

College Student

Graduate

Working Professional

2. Which areas are you most interested in?

Software Development

Data/AI

Cybersecurity

Cloud/DevOps

Business/Management

Design/Other

3. Which skills do you currently possess?

Programming

Communication

Problem Solving

Data Analysis

Web Development

Other

4. What type of recommendation would be most useful to you?

Career Role

Skills

Courses

Certifications

Projects

All of these

5. How important is personalized career guidance to you?

Very Important

Important

Moderately Important

Not Important

3) User Persona Creation

We can create personas for the main users of the AI Career Guidance and Skill Recommendation System.

Persona 1 — Student

Name: Riya

Role: College Student

Goal: Choose a suitable career path and understand what skills to develop.

Needs:

Personalized career suggestions

Skill-gap analysis

Learning roadmap

Course and project recommendations

Problem: Is unsure which career role matches her interests and current skills.

Persona 2 — Career Seeker

Name: Arjun

Role: Graduate / Job Seeker

Goal: Become job-ready for a target role and improve employability.

Needs:

Target-role recommendations

Required skill list

Resume-oriented skill guidance

Project and certification suggestions

Problem: Has basic knowledge but does not know which skills employers expect.

Persona 3 — Administrator / Career Counselor

Name: Neha

Role: Career Counselor / System Administrator

Goal: Support users and manage the career guidance platform.

Needs:

User insights

Recommendation reports

Skill and career data management

Access and system controls

Problem: A large number of users make personalized manual guidance difficult to scale.

4) Observation / Interview Simulation

For requirements gathering, we observe and interview students, graduates, career seekers, counselors, and administrators who would use or manage the career guidance system.

Interview Questions

Q1. How do you currently decide which career path to follow?

Answer: Career decisions are generally based on personal interests, academic performance, peer or family advice, online information, and perceived job opportunities.

Q2. What information is important for recommending a suitable career?

Answer: Interests, existing skills, educational background, preferred work areas, career goals, experience, and the skills demanded by target roles are important.

Q3. What should happen when the system identifies a skill gap?

Answer: The system should explain the missing skills and provide a prioritized learning roadmap with suitable courses, certifications, and practical projects.

Q4. Should every user receive the same career recommendation?

Answer: No. Recommendations should be personalized according to the user's profile, interests, current skills, goals, and target career.

Q5. Who should have access to the system?

Answer: Students and career seekers should be able to manage their profiles and view recommendations, while counselors and administrators should have access to relevant reports and management features.

5) System — Query

The proposed AI Career Guidance and Skill Recommendation System receives user information such as education, interests, skills, experience, and career preferences. It analyzes this information against career-role and skill data to generate personalized recommendations and identify skill gaps.

User Profile → AI Analysis → Career Matching → Skill-Gap Analysis → Recommendations → Learning Roadmap

The system can classify recommendations into:

High Match → Strongly suitable career role based on the user's profile.

Moderate Match → Suitable career role with some skills requiring improvement.

Low Match → Career role requiring significant skill development or a different preparation path.

Example:

A student has basic Python, problem-solving, and database skills and shows strong interest in Artificial Intelligence. The system compares the profile with relevant career roles, identifies missing skills such as machine learning, statistics, and model deployment, and recommends a structured learning path with courses and projects.

6) Stakeholder Identification

The major stakeholders of the AI Career Guidance and Skill Recommendation System are:

Student / Career Seeker

Career Counselor / Mentor

Institution / Placement Cell

System Administrator

Stakeholder Responsibilities

7) Requirement Elicitation Techniques

The following techniques are used to collect requirements for the AI Career Guidance and Skill Recommendation System:

1. Observation

Observe how students and job seekers currently search for career information, choose skills, and plan their learning.

2. Interview

Interview students, graduates, counselors, mentors, and placement stakeholders to understand their needs and challenges.

3. Questionnaire

Collect requirements from a larger group of users through structured questions about careers, skills, and learning preferences.

4. User Persona Creation

Create personas representing different user types to understand their goals, expectations, frustrations, and decision-making needs.

8) Consolidated Requirements

Functional Requirements

FR1 — User Registration/Login

The system should allow users to securely register, log in, and manage their profiles.

FR2 — Profile Management

The system should collect and maintain education, interests, skills, experience, goals, and preferences.

FR3 — Career Recommendation

The system should analyze the user's profile and recommend suitable career roles.

FR4 — Skill-Gap Analysis

The system should compare the user's current skills with the skills required for selected career roles.

FR5 — Skill Recommendation

The system should recommend relevant technical and soft skills to improve career readiness.

FR6 — Learning Roadmap

The system should provide a prioritized learning path based on identified skill gaps.

FR7 — Course/Certification Recommendation

The system should suggest relevant learning resources, courses, certifications, and practical projects.

FR8 — Career Exploration

Users should be able to view career-role information, required skills, and expected preparation.

FR9 — Reports

Counselors/administrators should be able to view useful recommendation and user-progress reports.

FR10 — User Management

The administrator should be able to add, remove, and manage users and system access.

Non-Functional Requirements

NFR1 — Security

The system should protect sensitive user profile and career-related information.

NFR2 — Performance

The system should generate recommendations within a reasonable response time.

NFR3 — Reliability

The system should provide consistent and dependable recommendations.

NFR4 — Scalability

The system should support a growing number of users, career roles, and skill records.

NFR5 — Availability

The system should be accessible whenever users require career guidance.

NFR6 — Usability

The interface should be simple, intuitive, and easy to understand.

NFR7 — Maintainability

The system should be easy to update as career roles, skills, and industry requirements change.

NFR8 — Recommendation Quality

The system should aim to provide relevant, explainable, and personalized recommendations while minimizing unsuitable suggestions.

9) Requirement Traceability

The following table connects each functional requirement with the relevant stakeholder.

Example Traceability

FR3 — Career Recommendation

Requirement: The system should recommend career roles that match the user's profile. Stakeholder: Student/Career Seeker, Counselor Source: Interview + Questionnaire Priority: High

FR4 — Skill-Gap Analysis

Requirement: The system should identify skills that the user needs to develop for a target career. Stakeholder: Student/Career Seeker, Counselor Source: Interview + Questionnaire Priority: High

10) Conclusion

The AI Career Guidance and Skill Recommendation System is designed to provide personalized and structured career guidance by combining user profiles, career requirements, skill analysis, and AI-based recommendations. Through observation, interviews, questionnaires, and user persona creation, the major requirements of different stakeholders were identified.

The consolidated functional and non-functional requirements provide a clear understanding of the proposed system. Requirement traceability ensures that each requirement is connected to the appropriate stakeholder and system functionality.

Overall, the system aims to reduce uncertainty in career decision-making, identify skill gaps, recommend relevant learning opportunities, and help students and career seekers build a focused roadmap toward their desired career.

Stakeholder	Requirement/Role
Student / Career Seeker	Create profile, explore career options, receive skill recommendations, and follow learning paths.
Career Counselor / Mentor	Review recommendations, guide users, and support career planning.
Institution / Placement Cell	Use aggregated insights to support training, employability, and placement preparation.
Administrator	Manage users, career/skill data, system access, and platform configuration.

ID	Requirement	Stakeholder
FR1	User Registration/Login	Student/Career Seeker, Administrator
FR2	Profile Management	Student/Career Seeker
FR3	Career Recommendation	Student/Career Seeker, Counselor
FR4	Skill-Gap Analysis	Student/Career Seeker, Counselor
FR5	Skill Recommendation	Student/Career Seeker, Counselor
FR6	Learning Roadmap	Student/Career Seeker
FR7	Course/Certification Recommendation	Student/Career Seeker, Counselor
FR8	Career Exploration	Student/Career Seeker
FR9	Reports	Counselor, Institution/Placement Cell
FR10	User Management	Administrator